AquaSentry Universal All-In-One Distribution

> **The definitive, consolidated distribution package for AquaSentry and AquaNet-V1.**
> Contains the full-stack web application, standalone neural network toolkit, C99 embedded firmware engine, AI agent tools, voltammetry datasets, and engineering documentation.

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1. Package Directory Layout

When unzipped on any computer, this package contains everything organized into clear, dedicated directories:

```
AquaSentry_Universal/
├── setup_and_run.sh            # 1-Click launcher for macOS & Linux (auto venv + pi
├── setup_and_run.bat           # 1-Click launcher for Windows (auto venv + pip + se
├── run_all_tests.sh            # 1-Click test runner for macOS & Linux (97 total te
├── run_all_tests.bat           # 1-Click test runner for Windows
├── main.py                     # Root universal command-line interface
├── requirements.txt            # Unified project dependencies
├── README.md                   # This documentation guide
├── README.pdf                  # Formatted PDF publication
│
├── AquaSentry_Platform/        # Universal Fullstack Web Application
│   ├── app/                    # FastAPI backend (endpoints, schemas, models, datab
│   ├── frontend/               # Interactive HTML5/CSS/JS dashboard with real-time
│   ├── bridge/                 # ADS1115 / ESP32 serial bridge and microcontroller
│   ├── tests/                  # 79 automated API, safety, and scientific integrati
│   ├── aquasentry.db           # Pre-configured SQLite database
│   ├── main.py                 # Platform-specific runner
│   └── run_tests.py            # Platform test runner with auto-venv detection
│
├── Neural_Network/             # Standalone AquaNet-V1 Neural Toolkit
│   ├── aquanet_classifier.py   # Pure-Python forward pass (112 synapses, no externa
│   ├── aquanet_neural.js       # Pure-JavaScript browser inference engine
│   ├── aquanet_v1.h            # C99 single-header microcontroller interface
│   ├── aquanet_v1.c            # C99 embedded inference implementation (zero dynami
│   ├── aquanet_v1_weights.json # Canonical 112 weight and bias floating-point matri
│   ├── asv_extractor.py        # Savitzky-Golay filtering & Faradaic peak current e
│   ├── test_aquanet_parity.py  # 18-point neural topology & strict parity test suit
```

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│   ├── main.py                 # Standalone ML CLI
│   └── AQUANET_NEURAL_NETWORK_COMPREHENSIVE_GUIDE.pdf
│
├── AI_Agent/                   # AI Water Quality Research Agent
│   ├── main.py                 # Multi-provider LLM agent (Gemini, Claude, GPT-4)
│   ├── tools.py                # Specialized search & toxicology query tools
│   └── requirements.txt        # Agent dependencies
│
├── Data/                       # Raw Analytical Datasets & Calibration Curves
│   ├── example_data.csv        # Standard 81-point ASV square-wave scan
│   ├── arsenic_voltammetry_example.csv
│   ├── voltammetry2_arsenic_example.csv
│   └── example_calibration.json# Slope and intercept calibration constants
│
├── Analysis/                   # Signal Processing & Trace Generators
│   ├── generate_arsenic_data.py# Synthetic voltammogram generation with noise model
│   └── mas_g1_arsenic_analysis.py # Historical baseline subtractor
│
└── Docs/                       # Complete Engineering Specifications & Blueprints
    ├── AQUANET_NEURAL_NETWORK_COMPREHENSIVE_GUIDE.pdf
    ├── CATAMARAN_PROTOTYPE_SPEC.pdf
    ├── PERSONAL_TANK_SENTRY_BLUEPRINT.pdf
    ├── AQUANET_RETRAINING_REPORT.pdf
    └── AQUANET_V1_STRESS_TEST_REPORT.pdf
```

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2. Quick Start on Any New Computer

macOS & Linux (1-Click)

In your terminal, navigate to the unzipped folder and run:

```
chmod +x setup_and_run.sh
./setup_and_run.sh
```

What this does automatically: 1. Detects system `python3`. 2. Creates an isolated local virtual environment (`.venv`). 3. Installs all required packages from `requirements.txt`. 4. Launches the AquaSentry web platform on `http://localhost:8000` and opens your browser.

Windows (1-Click)

Simply double-click:

```
setup_and_run.bat
```

or from the Command Prompt:

```
setup_and_run.bat
```

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3. Universal Command-Line Interface (`main.py`)

The root `main.py` provides immediate access to all subsystems:

```
# 1. Start the interactive web dashboard and REST API
python3 main.py --serve --port 8000

# 2. Run the complete automated test suite (all 97 tests)
python3 main.py --test

# 3. Analyze raw ASV voltammetric scan CSV and classify water safety
python3 main.py --csv Data/example_data.csv

# 4. Supply specific geochemical field parameters along with CSV
python3 main.py --csv Data/example_data.csv --ph 7.4 --ec 450 --turb 1.8 --temp 25.0

# 5. Run standalone AquaNet-V1 neural network evaluation scenarios
python3 main.py --neural

# 6. Display system architecture topology and versions
python3 main.py --version
```

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4. Running the Complete Test Suite

You can run all 97 tests across the entire platform in a single command:

```
./run_all_tests.sh
```

Test Coverage Breakdown: * **18 Standalone Neural Parity Tests** (`Neural_Network/test_aquanet_parity.py`): * Exact 23-node, 112-synapse topology verification. * Out-of-bounds (OOB) physical sensor fail-safes. * Multi-parameter WHO/BIS 10500 potability downgrades (pH, Turbidity, Conductivity). * Conservative precedence overrides for arsenic limits. * Numerical stability (softmax sum = 1.0, float64 scaling). * **79 Platform Integration Tests** (`AquaSentry_Platform/run_tests.py`): * REST API endpoints, health checks, reading ingestion, deduplication. * Signal processing (Savitzky-Golay filters, baseline tilt subtraction). * Chat policy prompt-injection filters & emergency toxicology gates. * Data export/import schema integrity.

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5. Microcontroller C99 Deployment

The embedded AquaNet-V1 inference engine is ANSI C99 compliant and has **zero dynamic memory allocations** (`malloc`), making it suitable for ESP32, STM32, Arduino, and ARM Cortex-M:

```
cd Neural_Network
clang -std=c99 -Wall -Wextra -Werror -c aquanet_v1.c
```

C99 API Usage:

```
#include "aquanet_v1.h"

AquaNetReading reading = {
    .arsenic_mg_l       = 0.0075f,
    .ph                 = 7.2f,
    .conductivity_us_cm = 400.0f,
    .turbidity_ntu      = 2.0f,
    .temperature_c      = 24.0f
};

AquaNetResult result;
aquanet_predict(&reading, &result);

// result.predicted_class: 0 = Safe, 1 = Caution, 2 = Unsafe, 3 = Hazardous
// result.confidence_pct:  Classification confidence (0.0 to 100.0%)
```

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6. Standards Compliance

- **WHO Guideline Limit**: 0.010 mg/L ($10\ \mu\text{g/L}$).
- **BIS 10500 Permissible Limit**: 0.050 mg/L ($50\ \mu\text{g/L}$).
- **Acute Hazard Limit**: $> 0.150\text{ mg/L}$ ($150\ \mu\text{g/L}$).
- **Method Limit of Detection (LOD)**: 0.020 mg/L ($20\ \mu\text{g/L}$).